

Effect Sizes in Cluster-Randomized Designs

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Multisite research designs involving cluster randomization are becoming increasingly important in educational and behavioral research. Researchers would like to compute effect size indexes based on the standardized mean difference to compare the results of cluster-randomized studies (and corresponding quasi-experiments) with other studies and to combine information across studies in meta-analyses. This article addresses the problem of defining effect sizes in multilevel designs and computing estimates of those effect sizes and their standard errors from information that is likely to be reported in journal articles. Three effect sizes are defined corresponding to different standardizations. Estimators of each effect size index are also presented along with their sampling distributions (including standard errors).

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Multisite studies are frequently used to evaluate the effects of educational treatments (e.g., interventions, products, or technologies). One common design assigns entire *sites* (often schools) to the same treatment group, with different sites assigned to different treatments. This design is often called a *cluster-randomized* design because sites such as schools correspond to statistical clusters. Several analysis strategies for cluster-randomized trials are possible. The simplest is to treat the clusters as units of analysis, that is, to compute mean scores on the outcome (and all other variables that may be involved in the analysis) and carry out the statistical analysis as if the site (cluster) means were the data. A more sophisticated alternative is to use a hierarchical linear modeling scheme with clusters as one level in the model (see, e.g., Raudenbush & Bryk, 2002). Many authors have commented on the problems of analyses of cluster-randomized trials (e.g., Donner & Klar, 2000; Klar & Donner, 2001; Murray, Varnell, & Blitstein, 2004; Raudenbush & Bryk, 2002).

Problems of representation of the results of cluster-randomized trials (and the corresponding quasi-experiments) in the form of effect sizes and combining them across studies in meta-analyses have received less attention. The problem of meta-analysis of cluster-randomized trials was considered by Rooney and

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Murray (1996), who called attention to the problem of effect size estimation in cluster-randomized trials and suggested that conventional estimates were not appropriate and their standard errors were incorrect. Donner and Klar (2002) suggested that corrections for the effects of clustering should be introduced in meta-analyses of cluster-randomized experiments. Laopaiboon (2003) reviewed the methods used in 25 published meta-analyses involving cluster-randomized experiments and found that only 3 used methods to account for clustering in their analysis. All three were meta-analyses of health care studies using binary outcomes. Of the six meta-analyses involving education, none used methods that addressed the impact of clustering.

This work was stimulated by problems faced by the U.S. Department of Education's What Works Clearinghouse, whose mission is to evaluate, to compare, and to synthesize evidence of effectiveness of educational programs, products, practices, and policies. The What Works Clearinghouse reviewers found that the majority of the high-quality studies they were examining involved assignment of treatment by clusters, which needed to be taken into account in computing an estimate of effect size and its uncertainty. This article has two purposes. One is to examine the problem of defining effect sizes for cluster-randomized trials. The second is to examine how to estimate these effect sizes and obtain standard errors for them from statistics that are typically given in reports of research (i.e., without a reanalysis of the raw data).

Model and Notation

Let Y_{ij}^T ($i = 1, \dots, m^T$; $j = 1, \dots, n_i^T$) and Y_{ij}^C ($i = 1, \dots, m^C$; $j = 1, \dots, n_i^C$) be the j th observation in the i th cluster in the treatment and control groups, respectively, so that there are m^T clusters in the treatment group and m^C clusters in the control group and a total of $M = m^T + m^C$ clusters with n observations each. Thus, the sample size is

$$N^T = \sum_{i=1}^{m^T} n_i^T$$

in the treatment group,

$$N^C = \sum_{i=1}^{m^C} n_i^C$$

in the control group, and the total sample size is $N = N^T + N^C$.

Let $\bar{Y}_{i\bullet}^T$ ($i = 1, \dots, m^T$) and $\bar{Y}_{i\bullet}^C$ ($i = 1, \dots, m^C$) be the means of the i th cluster in the treatment and control groups, respectively, and let $\bar{Y}_{\bullet\bullet}^T$ and $\bar{Y}_{\bullet\bullet}^C$ be the overall (grand) means in the treatment and control groups, respectively. Define the (pooled) within-cluster sample variance S_W^2 via

$$S_W^2 = \frac{\sum_{i=1}^{m^T} \sum_{j=1}^{n_i^T} (Y_{ij}^T - \bar{Y}_{i\bullet}^T)^2 + \sum_{i=1}^{m^C} \sum_{j=1}^{n_i^C} (Y_{ij}^C - \bar{Y}_{i\bullet}^C)^2}{N - M}$$

and the total pooled within-treatment group variance S_T^2 via

$$S_T^2 = \frac{\sum_{i=1}^{m^T} \sum_{j=1}^{n_i^T} (Y_{ij}^T - \bar{Y}_{\bullet\bullet}^T)^2 + \sum_{i=1}^{m^C} \sum_{j=1}^{n_i^C} (Y_{ij}^C - \bar{Y}_{\bullet\bullet}^C)^2}{N - 2}.$$

Let S_B be the pooled within-treatment groups standard deviation of the cluster means given by

$$S_B^2 = \frac{\sum_{i=1}^{m^T} (\bar{Y}_{i\bullet}^T - \bar{Y}_{\bullet\bullet}^T)^2 + \sum_{i=1}^{m^C} (\bar{Y}_{i\bullet}^C - \bar{Y}_{\bullet\bullet}^C)^2}{m^T + m^C - 2},$$

where $\bar{Y}_{\bullet\bullet}^T$ is the (unweighted) mean of the m^T cluster means in the treatment group and $\bar{Y}_{\bullet\bullet}^C$ is the (unweighted) mean of the m^C cluster means in the control group. That is,

$$\bar{Y}_{\bullet\bullet}^T = \frac{1}{m^T} \sum_{i=1}^{m^T} \bar{Y}_{i\bullet}^T$$

and

$$\bar{Y}_{\bullet\bullet}^C = \frac{1}{m^C} \sum_{i=1}^{m^C} \bar{Y}_{i\bullet}^C.$$

Note that when cluster sample sizes are unequal, $\bar{Y}_{\bullet\bullet}^T$ need not equal $\bar{Y}_{\bullet\bullet}^T$, the grand mean of the treatment group, and $\bar{Y}_{\bullet\bullet}^C$ need not equal $\bar{Y}_{\bullet\bullet}^C$, the grand mean of the control group. However, when cluster sample sizes are all equal, $\bar{Y}_{\bullet\bullet}^T = \bar{Y}_{\bullet\bullet}^T$ and $\bar{Y}_{\bullet\bullet}^C = \bar{Y}_{\bullet\bullet}^C$.

Suppose that observations within the treatment and control group clusters are normally distributed about cluster means μ_i^T and μ_i^C with a common within-cluster variance σ_W^2 . That is,

$$Y_{ij}^T \sim N(\mu_i^T, \sigma_W^2), i = 1, \dots, m^T; j = 1, \dots, n_i^T$$

and

$$Y_{ij}^C \sim N(\mu_i^C, \sigma_W^2), i = 1, \dots, m^C; j = 1, \dots, n_i^C.$$

Suppose further that the clusters are random effects (e.g., they are considered a sample from a population of clusters) so that the cluster means themselves have a normal sampling distribution with means $\mu_{\bullet}^T$ and $\mu_{\bullet}^C$ and common variance σ_B^2 . That is,

$$\mu_i^T \sim N(\mu_{\bullet}^T, \sigma_B^2), i = 1, \dots, m^T$$

and

$$\mu_i^C \sim N(\mu_{\bullet}^C, \sigma_B^2), i = 1, \dots, m^C.$$

Note that in this formulation, σ_B^2 represents true variation of the population means of clusters over and above the variation in sample means that would be expected from variation in the sampling of observations into clusters.

These assumptions correspond to the usual assumptions that would be made in the analysis of a multisite trial by a hierarchical linear models analysis, an analysis of variance (with treatment as a fixed effect and cluster as a nested random effect), or a t test using the cluster means in the treatment and control group as the unit of analysis.

In principle, there are three different within-treatment group standard deviations, σ_B , σ_W , and σ_T , the latter defined by

$$\sigma_T^2 = \sigma_B^2 + \sigma_W^2.$$

In most educational data when clusters are schools, σ_B^2 is considerably smaller than σ_W^2 . Obviously, if the between-cluster variance σ_B^2 is small, then σ_T^2 will be very similar to σ_W^2 .

A parameter that summarizes the relationship between the three variances is called the intraclass correlation ρ , which is defined by

$$\rho = \frac{\sigma_B^2}{\sigma_B^2 + \sigma_W^2} = \frac{\sigma_B^2}{\sigma_T^2}. \quad (1)$$

The intraclass correlation ρ can be used to obtain one of these variances from any of the others, because $\sigma_W^2 = (1 - \rho)\sigma_B^2/\rho$, $\sigma_W^2 = (1 - \rho)\sigma_T^2$, and $\sigma_B^2 = \rho\sigma_T^2$.

Effect Sizes

The effect sizes typically used in educational and psychological research are standardized mean differences, defined as the ratio of a difference between treatment and control group means to a standard deviation. In single-site designs or designs where there is no statistical clustering, the notion of standardized mean difference is often unambiguous: There is only one possibility. In multisite designs such as cluster-randomized trials, there are several possible standardized mean differences. This section clarifies the possibilities.

The three possibilities for the standard deviation lead to different possible definitions for the population effect size in this clustered design. The choice of one of these effect sizes should be determined on the basis of the inference of interest to the researcher. If the effect size measures are to be used in meta-analysis, an important inference goal may be to estimate parameters that are comparable with those that can be estimated in other studies. In such cases, the standard deviation may be chosen to be the same kind of standard deviation used in the effect sizes of other studies with which this study will be compared. The focus here is on three effect sizes that seem likely to be the most useful (meaning the most widely used).

If $\sigma_W \neq 0$ (and hence $\rho \neq 1$), one effect size parameter has the form

$$\delta_W = \frac{\mu_{\bullet}^T - \mu_{\bullet}^C}{\sigma_W}. \quad (2)$$

This effect size might be of interest, for example, in a meta-analysis where the other studies with which the current study is compared are typically single-site studies. In such studies, δ_W may (implicitly) be the effect size estimated, and hence, δ_W might be the effect size most comparable with that in other studies.

A second effect size parameter is of the form

$$\delta_T = \frac{\mu_{\bullet}^T - \mu_{\bullet}^C}{\sigma_T}. \quad (3)$$

This effect size might be of interest in a meta-analysis where the other studies are multisite studies or studies that sample from a broader population but do not include clusters in the sampling design (this would typically imply that they used an individual, rather than a cluster, assignment strategy). In such cases, δ_T might be the most comparable with the effect sizes in other studies.

If $\sigma_B \neq 0$ (and hence $\rho \neq 0$), a third possible effect size parameter would be

$$\delta_B = \frac{\mu_{\bullet}^T - \mu_{\bullet}^C}{\sigma_B}. \quad (4)$$

This effect size is less likely to be of general interest, but it might be of interest in cases where the treatment effect is defined at the level of clusters and the individual observations are of interest because the average defines an aggregate property. The effect size δ_B may also be of interest in a meta-analysis where the studies being compared are typically multisite studies that have been analyzed by using cluster means as the unit of analysis.

Note, however, that although δ_W and δ_T will often be similar in magnitude, δ_B will typically be much larger (in the same population) than either δ_W or δ_T , because σ_B is typically considerably smaller than σ_W and σ_T . Note also that if all of the effect sizes are defined (i.e., if $0 < \rho < 1$), and ρ is known, any one of

these effect sizes may be obtained from any of the others. In particular, both δ_W and δ_T can be obtained from δ_B and ρ because

$$\delta_W = \delta_B \sqrt{\frac{\rho}{1-\rho}} = \frac{\delta_T}{\sqrt{1-\rho}} \quad (5)$$

and

$$\delta_T = \delta_B \sqrt{\rho} = \delta_W \sqrt{1-\rho}. \quad (6)$$

Estimates of Effect Sizes: Equal Cluster Sample Sizes

Although it is easy to define different effect sizes in multilevel (e.g., clustered) designs, the nested variance structure makes estimation somewhat less intuitive than in single-level designs. In this section, we present estimates of the effect sizes and their approximate sampling distributions when the cluster sample sizes are equal to n , that is, when $n_i^T = n$, $i = 1, \dots, m^T$ and $n_i^C = n$, $i = 1, \dots, m^C$. In this case, $N^T = nm^T$, $N^C = nm^C$, and $N = N^T + N^C = n(m^T + m^C) = nM$. Derivations and the details of the small sample distribution of the effect size estimators are given in the appendix.

Results are presented explicitly for the case of equal cluster sample sizes for two reasons. The first reason is that most designs attempt to achieve equal cluster sample sizes, but specific (realized) cluster sample sizes are rarely reported, so that the equal sample size formulas will be of most practical use. The second reason is that the results become considerably more complicated when cluster sample sizes are unequal—sufficiently complicated that it is difficult to obtain much insight from examining the formulas in the case of the unequal cluster sample size.

Estimation of δ_W

Start with estimation of δ_W , which is the most straightforward. If $\rho \neq 1$, the estimate

$$d_W = \frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{S_W} \quad (7)$$

is a consistent estimator of δ_W . The estimator d_W is approximately normally distributed about δ_W with variance

$$V\{d_W\} = \left(\frac{N^T + N^C}{N^T N^C} \right) \left(\frac{1 + (n-1)\rho}{1-\rho} \right) + \frac{\delta_W^2}{2(N-M)}. \quad (8)$$

An estimate v_W of the variance of d_W can be computed by substituting the consistent estimate d_W for δ_W in Equation 8 above. Note that the presence of the factor $(1 - \rho)$ in the denominator of the first term is possible because δ_W is defined only if $\rho \neq 1$.

Note that if $\rho = 0$ and there is no clustering, Equation 8 reduces to the variance of a mean difference divided by a standard deviation with $(N - M)$ degrees of freedom (see Hedges, 1981). The leading term of the variance in Equation 8 arises from uncertainty in the mean difference. Note that it is $[1 + (n - 1)\rho]/(1 - \rho)$ as large as would be expected if there were no clustering in the sample (i.e., if $\rho = 0$). Thus $[1 + (n - 1)\rho]/(1 - \rho)$ is a kind of variance inflation factor for the variance of the effect size estimate d_W .

Estimation of δ_B

Estimation of the other δ_B and δ_T is less intuitive than that of δ_W . For example, one might expect that

$$\frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{S_B}$$

would be that natural estimator of δ_B , but this is not the case. The reason is that S_B is not a pure estimate of σ_B because it is inflated by the within-cluster variability. In particular, the expected value of S_B^2 is

$$\sigma_B^2 + \frac{\sigma_W^2}{n}.$$

If an estimate S_W^2 of the (average) within-cluster variance is reported, then it is possible to obtain an estimate $\hat{\sigma}_B^2$ of σ_B^2 by subtraction, namely,

$$\hat{\sigma}_B^2 = S_B^2 - \frac{S_W^2}{n},$$

whenever this quantity is nonnegative and zero otherwise. Whenever $\hat{\sigma}_B^2$ is nonzero, one estimate of δ_B is therefore

$$d_{B1} = \frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{\hat{\sigma}_B}. \quad (9)$$

Whenever δ_B is defined (i.e., when $\rho \neq 0$), d_{B1} is normally distributed in large samples with variance

$$V\{d_{B1}\} = \left(\frac{m^T + m^C}{m^T m^C} \right) \left(\frac{1 + (n - 1)\rho}{n\rho} \right) + \left[\frac{[1 + (n - 1)\rho]^2}{2(M - 2)n^2\rho^2} + \frac{(1 - \rho)^2}{2n^2(N - M)\rho^2} \right] \delta_B^2. \quad (10)$$

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An estimate v_{B1} of the variance of d_{B1} can be computed by substituting the consistent estimate d_{B1} for δ_B in Equation 10 above. The estimate d_{B1} has the virtue that it can be computed without an external estimate of ρ . Note that the presence of ρ in the denominators of the variance terms is possible because δ_B is only defined if $\rho \neq 0$.

Alternatively (again assuming that $\rho \neq 0$ so that δ_B is defined), the intraclass correlation may be used to obtain an estimate of δ_B using S_B . A direct argument shows that

$$d_{B2} = \frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{S_B} \sqrt{\frac{1 + (n-1)\rho}{n\rho}} \quad (11)$$

is a consistent estimate of δ_B . This estimate is normally distributed in large samples with variance

$$V\{d_{B2}\} = \left(\frac{m^T + m^C}{m^T m^C} \right) \left(\frac{1 + (n-1)\rho}{n\rho} \right) + \frac{[1 + (n-1)\rho]\delta_B^2}{2n\rho(M-2)}. \quad (12)$$

An estimate v_{B2} of the variance of d_{B2} can be computed by substituting the consistent estimate d_{B2} for δ_B in Equation 12 above. Note that the presence of ρ in the denominators of the variance terms is possible because δ_B is only defined if $\rho \neq 0$.

The variance in Equation 12 is $[1 + (n-1)\rho]/n\rho$ as large as the variance of the standardized mean difference computed from an analysis using cluster means as the unit of analysis, that is, applying the usual formula for the variance of the standardized difference between the means of the cluster means in the treatment and control groups. Thus, $[1 + (n-1)\rho]/n\rho$ is a kind of variance inflation factor for the variance of effect size estimates like d_B compared to the alternative effect size estimate based on cluster means.

Estimation of δ_T

An estimate of δ_T can also be obtained in either of two ways. If the pooled within-treatment groups variance of the cluster means S_B^2 and the pooled within-cluster variance S_W^2 are both known, then an estimate of (σ_T) can be constructed as

$$\hat{\sigma}_T = \sqrt{S_B^2 + \left(\frac{n-1}{n} \right) S_W^2}.$$

The estimator

$$d_{T1} = \frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{\hat{\sigma}_T} \quad (13)$$

is a consistent estimator of δ_T . This estimate is normally distributed about δ_T in large samples with variance

$$V\{d_{T1}\} = \left(\frac{N^T + N^C}{N^T N^C} \right) (1 + (n-1)\rho) + \left[\frac{[1 + (n-1)\rho]^2}{2n^2(M-2)} + \frac{(n-1)^2(1-\rho)^2}{2n^2(N-M)} \right] \delta_T^2. \quad (14)$$

An estimate v_{T1} of the variance of d_{T1} can be computed by substituting the consistent estimate d_{T1} for δ_T in Equation 14 above. The estimate d_{T1} has the virtue that it can be computed without an external estimate of ρ .

Alternatively, the intraclass correlation and S_T may be used to obtain an estimate of δ_T . A direct argument shows that a consistent estimator of δ_T is

$$d_{T2} = \left(\frac{\bar{Y}_{..}^T - \bar{Y}_{..}^C}{S_T} \right) \sqrt{1 - \frac{2(n-1)\rho}{N-2}}. \quad (15)$$

It is normally distributed in large samples with variance

$$V\{d_{T2}\} = \left(\frac{N^T + N^C}{N^T N^C} \right) (1 + (n-1)\rho) + \delta_T^2 \left(\frac{(N-2)(1-\rho)^2 + n(N-2n)\rho^2 + 2(N-2n)\rho(1-\rho)}{2(N-2)[(N-2) - 2(n-1)\rho]} \right). \quad (16)$$

An estimate v_{T2} of the variance of d_{T2} can be computed by substituting the consistent estimate d_{T2} for δ_T in Equation 16 above. Note that if $\rho = 0$ and there is no clustering, d_{T2} reduces to the conventional standardized mean difference and Equation 16 reduces to the usual expression for the variance of the standardized mean difference (see Hedges, 1981).

The leading term of the variance in Equations 14 and 16 arises from uncertainty in the mean difference. Note that this leading term is $[1 + (n-1)\rho]$, as large as would be expected if there were no clustering in the sample (i.e., if $\rho = 0$). The expression $[1 + (n-1)\rho]$ is the variance inflation factor mentioned by Donner, Birkett, and Buck (1981) and the design effect mentioned by Kish (1965) for the variance of means in clustered samples and also corresponds to a variance inflation factor for the effect size estimates like d_{T1} and d_{T2} .

Confidence Intervals for δ_W , δ_B , and δ_T

The results in this article can also be used to compute confidence intervals for effect sizes. If δ is any one of the effect sizes mentioned, d is a corresponding estimate, and v_d is the estimated variance of d , then a $100(1 - \alpha)\%$ confidence interval for δ based on d and v_d is given by

$$d - c_{\alpha/2} v_d \leq \delta \leq d + c_{\alpha/2} v_d, \quad (17)$$

where $c_{\alpha/2}$ is the $100(1 - \alpha/2)$ percentage point of the standard normal distribution (e.g., 1.96 for $\alpha/2 = 0.05/2 = 0.025$).

Estimates of Effect Size: Unequal Cluster Sample Sizes

When cluster sample sizes are unequal, expressions for the effect size estimators and their variances are considerably more complex. This section gives estimates of the effect sizes and their sampling distributions when cluster sample sizes are not equal. These expressions may be of use when cluster sample sizes are unequal and are reported explicitly. They also provide some insight about what single “compromise” sample size might give the most accurate results (e.g., in computing the variances of estimates) when substituted into the equal sample size formulas.

Estimation of δ_W

When $\rho \neq 1$, the estimator d_W of δ_W is the same as in the case of equal cluster sample sizes, but the variance of the estimator is given by

$$V\{d_W\} = \left(\frac{N^T + N^C}{N^T N^C} \right) \left(\frac{1 + (\tilde{n} - 1)\rho}{1 - \rho} \right) + \frac{\delta_W^2}{2(N - M)}, \quad (18)$$

where

$$\tilde{n} = \frac{N^C \sum_{i=1}^{m^T} (n_i^T)^2}{N^T N} + \frac{N^T \sum_{i=1}^{m^C} (n_i^C)^2}{N^C N}.$$

When all the n_i^T and n_i^C are equal to n , $\tilde{n} = n$ and Equation 18 reduces to Equation 8.

Estimation of δ_T

The form of the estimator d_{T2} is somewhat different when cluster sample sizes are unequal. In this case, the estimator becomes

$$d_{T2} = \left(\frac{\bar{Y}_{..}^T - \bar{Y}_{..}^C}{S_T} \right) \sqrt{1 - \rho \left(\frac{(N - n_U^T m^T - n_U^C m^C) + n_U^T + n_U^C - 2}{N - 2} \right)}, \quad (19)$$

where

$$n_U^T = \frac{(N^T)^2 - \sum_{i=1}^{m^T} (n_i^T)^2}{N^T (m^T - 1)}$$

and

$$n_U^C = \frac{(N^C)^2 - \sum_{i=1}^{m^C} (n_i^C)^2}{N^C(m^C - 1)}.$$

When all the n_i^T and n_i^C are equal to n , $n_U^T = n_U^C = n$, and Equation 19 reduces to Equation 15.

The variance of d_{T2} is somewhat more complex. It is given by

$$V\{d_{T2}\} = \left(\frac{N^T + N^C}{N^T N^C} \right) (1 + (\tilde{n} - 1)\rho) + \frac{\left[(N - 2)(1 - \rho)^2 + A\rho^2 + 2B\rho(1 - \rho) \right] \delta^2}{2(N - 2)[(N - 2) - \rho(N - 2 - B)]}, \quad (20)$$

where the auxiliary constants A and B are defined by $A = A^T + A^C$,

$$A^T = \frac{(N^T)^2 \sum_{i=1}^{m^T} (n_i^T)^2 + \left(\sum_{i=1}^{m^T} (n_i^T)^2 \right)^2 - 2N^T \sum_{i=1}^{m^T} (n_i^T)^3}{(N^T)^2},$$

$$A^C = \frac{(N^C)^2 \sum_{i=1}^{m^C} (n_i^C)^2 + \left(\sum_{i=1}^{m^C} (n_i^C)^2 \right)^2 - 2N^C \sum_{i=1}^{m^C} (n_i^C)^3}{(N^C)^2},$$

and

$$B = n_U^T(m^T - 1) + n_U^C(m^C - 1).$$

When n_i^T and n_i^C are equal to n , $A = n(N - 2n)$, $n_U^T = n_U^C = n$, and $B = (N - 2n)$ so that Equation 20 reduces to Equation 16. These expressions suggest that if cluster sample sizes are unequal, substituting the average of n_U^T and n_U^C into Equation 15 and Equation 16 would give results that are quite close to the exact values.

Estimation of δ_B

There is more than one way to generalize the estimator d_{B2} to the case of unequal cluster sample sizes. One possibility is to use the means of the cluster means in the treatment and control groups in the numerator and standard deviation of the cluster means in the denominator. This corresponds to using the cluster means as the unit of analysis. Another possibility for the numerator is to use the grand means in the treatment and control groups. Similarly, there are multiple possibilities for the denominator, such as some function of the mean square between groups. When cluster sample sizes are identical, then all these approaches

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are equivalent in the sense that the effect size estimates are identical. When the cluster sample sizes are not identical, the resulting estimators are not the same. Because the use of cluster means as the unit of analysis is a common approach, the means and standard deviations of cluster means are most likely to be available, and hence the sampling distribution of the effect size estimate is given based on the standard deviation of the cluster means.

When $\rho \neq 0$, an estimator of δ_B , which is a generalization of Equation 11, is given by

$$d_B = \left(\frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{S_B} \right) \sqrt{\frac{1 + (\bar{n}_B - 1)\rho}{\bar{n}_B \rho}}, \quad (21)$$

where

$$\bar{n}_B = \left(\frac{(m^T - 1)\bar{n}_I^T + (m^C - 1)\bar{n}_I^C}{M - 2} \right)^{-1},$$

$$\bar{n}_I^T = \frac{1}{m^T} \sum_{i=1}^{m^T} (1/n_i^T),$$

and

$$\bar{n}_I^C = \frac{1}{m^C} \sum_{i=1}^{m^C} (1/n_i^C).$$

The variance of d_B is approximately

$$V\{d_B\} = \left(\frac{m^T + m^C}{m^T m^C} \right) \left(\frac{1 + (\bar{n}_B - 1)\rho}{\bar{n}_B \rho} \right) + \frac{\bar{n}_B C \delta_B^2}{2(M - 2)^2 \rho [1 + (\bar{n}_B - 1)\rho]}, \quad (22)$$

where

$$\bar{n}_B = \left(\frac{m^C \bar{n}_I^T + m^T \bar{n}_I^C}{M} \right)^{-1},$$

$$C = (M - 2)\rho^2 + 2[(m^T - 1)\bar{n}_I^T + (m^C - 1)\bar{n}_I^C]\rho(1 - \rho)$$

$$+ [(m^T - 2)\bar{n}_I^{T2} + (m^C - 2)\bar{n}_I^{C2} + (\bar{n}_I^T)^2 + (\bar{n}_I^C)^2](1 - \rho)^2,$$

$$\bar{n}_I^{T2} = \frac{1}{m^T} \sum_{i=1}^{m^T} (1/n_i^T)^2,$$

and

$$\bar{n}_I^{C2} = \frac{1}{m^C} \sum_{i=1}^{m^C} (1/n_i^C)^2.$$

Note that when the n_i^T and n_i^C are all equal to n , $\bar{n}_B = n$, $\tilde{n}_B = n$, $\bar{n}_I^T = \bar{n}_I^C = 1/n$, $\bar{n}_I^{T2} = \bar{n}_I^{C2} = 1/n^2$, and

$$C = \frac{(M-2)[1 + (n-1)\rho]^2}{n^2}$$

so that Equation 21 reduces to Equation 11 and Equation 22 reduces to Equation 12. These expressions suggest that when cluster sample sizes are unequal, substituting $\bar{n}_B$ for n in Equations 11 and 12 would give results that are very close to the exact values.

Applications in Meta-Analysis

The statistical results in this article should be useful in deciding what effect sizes are desirable in a cluster-randomized experiment. They should also be useful for finding ways to compute effect size estimates and their variances from data that may be reported. In the sections that follow, applications are illustrated in some examples.

Intraclass correlations needed for the methods described in this article are often not reported. However, because plausible values of ρ are essential for power and sample size computations in planning cluster-randomized experiments, there have been systematic efforts to obtain information about reasonable values of ρ in realistic situations. Some information about reasonable values of ρ comes from cluster-randomized trials that have been conducted. For example, Murray and Blitstein (2003) reported a summary of intraclass correlations obtained from 17 articles reporting cluster-randomized trials in psychology and public health, and Murray et al. (2004) gave references to 14 very recent studies that provide data on intraclass correlations for health-related outcomes. Other information on reasonable values of ρ comes from sample surveys that use clustered sampling designs. For example, Guilliford, Ukoumunne, and Chinn (1999) and Verma and Lee (1996) presented values of intraclass correlations based on surveys of health outcomes. Hedges and Hedberg (2007) presented a compendium of several hundred intraclass correlations for academic achievement computed from national probability samples at various grade levels. This later compendium provides both national values for intraclass correlations and values for regions of the country and subsets of regions differing in level of urbanicity.

Computing Effect Sizes When Individuals Are the Unit of Analysis

The results given in this article can be used to produce effect size estimates and their variances from studies that incorrectly analyze cluster-randomized trials as if individuals were randomized. The required means, standard deviations, and sample sizes can usually be extracted from what may be reported.

Suppose it is decided that the effect size δ_T is appropriate because most other studies both assign and sample individually from a clustered population. Suppose that the data are analyzed by ignoring clustering; then the test statistic is likely to be either

$$t = \sqrt{\frac{N^T N^C}{N^T + N^C}} \left(\frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{S_T} \right)$$

or

$$F = \left(\frac{N^T N^C}{N^T + N^C} \right) \left(\frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{S_T} \right)^2.$$

Either can be solved for

$$\left(\frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{S_T} \right),$$

which can then be inserted into Equation 15 along with ρ to obtain d_{T2} . This estimate (d_{T2}) of δ_T can then be inserted into Equation 16 to obtain v_{T2} , an estimate of the variance of d_{T2} .

Alternatively, suppose it is decided that the effect size δ_W is appropriate because most other studies involve only a single site. Begin by computing d_{T2} and v_{T2} as before. Because an estimate of δ_W , not δ_T , is wanted, use the fact given in Equation 5 that

$$\delta_W = \frac{\delta_T}{\sqrt{1 - \rho}},$$

and therefore

$$\frac{d_{T2}}{\sqrt{1 - \rho}} \tag{23}$$

is an estimate of δ_W with a variance of

$$\frac{v_{T2}}{1 - \rho}. \tag{24}$$

Example

An evaluation of the connected mathematics curriculum reported by Ridgway, Zawgewski, Hoover, and Lambdin (2002) compared the achievement of $m^T = 18$ classrooms of sixth-grade students who used connected mathematics with that of $m^C = 9$ classrooms in a comparison group that did not use connected mathematics. In this quasi-experimental design, the clusters were classrooms. The cluster sizes were not identical, but the average cluster size in the treatment groups was $N^T/m^T = 338/18 = 18.8$ and $N^C/m^C = 162/18 = 9$ in the control group. The exact sizes of all the clusters were not reported, but here the cluster sizes are treated as if they were equal and $n = 18$ chosen as a slightly conservative sample size. The mean difference between treatment and control groups is $\bar{Y}_{..}^T - \bar{Y}_{..}^C = 1.9$, and the pooled within-groups standard deviation $S_T = 12.37$. This evaluation involved sites in all regions of the country, and it was intended to be nationally representative. Ridgway et al. did not give an estimate of the intraclass correlation based on their sample. Hedges and Hedberg (2007) provided an estimate of the Grade 6 intraclass correlation in mathematics achievement for the nation as a whole (based on a national probability sample) of 0.264 with a standard error of 0.019. For this example, it is assumed that the intraclass correlation is identical to that estimate, namely, $\rho = 0.264$.

Suppose that the analysis ignored clustering and compared the mean of all of the students in the treatment group with the mean of all of the students in the control group. This leads to a value of the standardized mean difference of

$$\frac{\bar{Y}_{..}^T - \bar{Y}_{..}^C}{S_T} = 0.1536,$$

which is not an estimate of any of the three effect sizes considered here. If an estimate of the effect size δ_T is desired and an intraclass correlation of $\rho = 0.264$ was imputed, then Equation 15 is used to obtain

$$d_{T2} = (0.1536)(0.9907) = 0.1522.$$

The effect size estimate is very close to the original standardized mean difference because the amount of clustering in this case is rather small. However, even this small amount of clustering has a substantial effect on the variance of the effect size estimate. The variance of the standardized mean difference ignoring clustering is

$$\frac{324 + 162}{324 \cdot 162} + \frac{0.1531^2}{2(324 + 162 - 2)} = 0.009259.$$

However, computing the variance of d_{T2} using Equation 16 with $\rho = 0.264$ obtains a variance estimate of 0.050865, which is 549% of the variance ignoring clustering. A 95% confidence interval for δ_T is given by

$$-0.2899 = 0.1522 - 1.96\sqrt{0.050865} \leq \delta_T \leq 0.1522 + 1.96\sqrt{0.050865} = 0.5942.$$

If clustering had been ignored, the confidence interval for the population effect size would have been -0.0350 to 0.3422 .

If one wanted to estimate δ_W , then an estimate of δ_W given by Expression 23 is

$$\frac{0.1522}{\sqrt{1-0.264}} = 0.1774,$$

with variance given by Expression 24 as

$$0.050865/(1-0.264) = 0.06911,$$

and a 95% confidence interval for δ_W based on Equation 17 would be

$$-0.3379 = 0.1774 - 1.96\sqrt{0.06911} \leq \delta_W \leq 0.1774 + 1.96\sqrt{0.06911} = 0.6926.$$

Computing Effect Sizes When Clusters Are the Unit of Analysis

The results given in this article can also be used to obtain different effect size estimates when the data have been analyzed with the cluster mean as the unit of analysis. In such cases, the researcher might report a t test or an analysis of variance carried out on cluster means, but here the wish is to estimate δ_T . In this case, the test statistic reported will either be

$$t = \sqrt{\frac{m^T m^C}{m^T + m^C}} \left(\frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{S_B} \right)$$

or

$$F = \left(\frac{m^T m^C}{m^T + m^C} \right) \left(\frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{S_B} \right)^2.$$

Either can be solved for

$$\left(\frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{S_B} \right),$$

which can then be inserted into Equation 11 along with ρ to obtain d_{B2} . This estimate of d_{B2} can then be inserted into Equation 12 to obtain v_{B2} , an estimate of the variance of d_{B2} . Because an estimate of δ_T , not δ_B , is wanted, use the fact given in Equation 6 that

$$\delta_T = \delta_B \sqrt{\rho},$$

and therefore

$$d_{B2}\sqrt{\rho} \quad (25)$$

is an estimate of δ_T with a variance of

$$\rho v_{B2}. \quad (26)$$

Alternatively, suppose it is decided that the effect size δ_W is the desired effect size. Begin by computing d_{B2} and v_{B2} as before. Because an estimate of δ_W , not δ_B , is wanted, use the fact given in Equation 5 that

$$\delta_W = \delta_B \sqrt{\frac{\rho}{1-\rho}},$$

and therefore

$$d_{B2}\sqrt{\frac{\rho}{1-\rho}} \quad (27)$$

is an estimate of δ_W with a variance of

$$\frac{\rho v_{B2}}{1-\rho}. \quad (28)$$

Example

An evaluation of University of Chicago School Mathematics Project (UCSMP) Geometry reported by Senk (2002) compared the results of $m^T = 8$ classrooms using UCSMP Geometry curriculum with $m^C = 8$ comparison classrooms that did not use the UCSMP curriculum. In this quasi-experimental design, clusters (classrooms) were the unit of analysis. The cluster sizes were not identical, but the average cluster size in the treatment group was $N^T/m^T = 139/8 = 17.4$ and $N^C/m^C = 115/8 = 14.4$ in the comparison group. The exact sizes of all the clusters were reported, but here the cluster sizes are treated as if they were equal and $n = 15$ chosen as a slightly conservative compromise sample size. The mean difference between treatment and control groups is $\bar{Y}_{..}^T - Y_{..}^C = -0.84$, and the pooled within-groups standard deviation $S_B = 2.034$. This evaluation involved sites in several regions of the country, and it was intended to be nationally representative. Senk did not give an estimate of the intraclass correlation based on her sample. Hedges and Hedberg (2007) provided an estimate of the intraclass correlation in mathematics achievement in Grade 10 for the nation as a whole (based on a national probability sample) of .234 with a standard error of 0.010. For this example, we assume that the intraclass correlation is identical to that estimate, namely, $\rho = 0.234$.

These values lead to a value of the standardized mean difference of

$$\frac{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C}{S_B} = -0.4130,$$

which is not an estimate of any of the three effect sizes considered here. If an estimate of the effect size δ_B is desired and an intraclass correlation of $\rho = 0.234$ was imputed, then Equation 11 is used to obtain

$$d_{B2} = (-0.4130)(1.2649) = -0.4558,$$

which is 26% larger than the unadjusted standardized mean difference. The variance of the standardized mean difference ignoring clustering is

$$\frac{8+8}{8 \times 8} + \frac{(-0.4130)^2}{2(8+8-2)} = 0.2577.$$

However, computing the variance of d_{B2} using Equation 12 with $\rho = 0.234$ obtains a variance estimate of 0.3239, which is about 60% larger than the variance computed ignoring clustering. A 95% confidence interval for δ_B based on Equation 17 is

$$-1.5712 = -0.4558 - 1.96\sqrt{0.3239} \leq \delta_W \leq -0.4558 + 1.96\sqrt{0.3239} = 0.6596.$$

If one wanted to estimate δ_T , using Expression 25 with $\rho = 0.234$ obtains

$$-0.4558\sqrt{0.234} = -0.2205$$

as an estimate of δ_T with a variance given by Expression 26 as $0.3239(0.234) = 0.0758$.

If one wanted to estimate δ_W , using Expression 27 with $\rho = 0.234$, obtains

$$-0.4558\sqrt{\frac{0.234}{1-0.234}} = -0.2519$$

as an estimate of δ_W with a variance given by Expression 28 as $(0.3239)[0.234/(1-0.234)] = 0.0989$.

The report of this study (Senk, 2002) gives the sample sizes for each cluster, which range from 5 to 25 and therefore are not nearly all equal. Because all the individual cluster sample sizes are given, it is possible to compute d_B and its variance using the formulas for unequal sample sizes. Using the data in Table 1,

TABLE 1

Data From the Evaluation of UCSMP Geometry Second Edition: HSST Geometry Test

UCSMP			Comparison		
<i>n</i>	<i>M</i>	<i>SD</i>	<i>n</i>	<i>M</i>	<i>SD</i>
9	34.4	10.1	7	39.3	13.7
5	29.0	8.9	9	36.7	14.1
22	50.3	12.7	13	43.5	12.7
20	48.3	9.3	17	42.2	12.3
20	46.8	15.0	19	48.4	14.8
17	47.5	10.7	15	49.7	10.2
25	40.4	10.0	14	38.6	15.0
21	33.8	10.8	21	38.8	17.1

Note: These data are from Senk (2002). UCSMP = University of Chicago School Mathematics Project; HSST = High School Subjects Test.

compute $\bar{n}_B = 12.997$, and using Equation 21, compute $d_B = (-0.4139)(1.1189) = -0.4621$. Also compute $\bar{n}_B = 12.997$, $\bar{n}_I^{T^2} = 0.008342$, $\bar{n}_I^{C^2} = 0.007089$, and $A = 0.4185$ so that the estimate of v_B using Equation 22 is 0.2820. Comparing the values of d_B (-0.4558 vs. -0.4621) and estimates of the variance (0.3239 vs. 0.2820) assuming equal cluster sample sizes with those using the exact cluster sample sizes, one sees that even with these large discrepancies among sample sizes, the values of d_B and the variance estimates assuming equal cluster sample sizes are within 15% of the actual values. If the value $n = 13$ had been used for the (common) cluster sample size (approximately the value of $\bar{n}_B$ or $\bar{n}_B$), the results using the equal sample size formulas would have been very close to the exact values.

Conclusion

This article has provided definitions of three different effect sizes that can be estimated in studies using cluster randomization. Alternative methods of estimation are provided for each effect size, and the sampling variances are also given. The sampling distribution of each estimator is shown to be a constant times a noncentral t distribution, and simple normal approximations are given in each case. Because these approximations have been extensively studied in the context of simpler effect size estimates and power analysis, there is reason to believe that they are reasonably accurate unless sample sizes are quite small (which is unlikely in cluster-randomized designs). Simulation studies (not reported here) evaluating the accuracy of these approximations confirm expectations.

The analytic work shows that clustering can have a substantial effect on the variance of effect size estimates in cluster-randomized designs. The example provided illustrates that small amounts of clustering can have a large effect on

the variance of effect sizes, even if the effect on the expected value of the estimates is modest. The results given in this article can be used to estimate the effect sizes (and their variances) in cluster-randomized trials that have been improperly analyzed by ignoring clustering, provided an intraclass correlation is known or can be imputed. The effect size estimates can then be used in meta-analyses along with any other effect size estimates of the same conceptual parameter, using the variances of the estimates to compute weights in the usual way.

The results given in this article require that a value of the intraclass correlation parameter ρ be known or imputed for sensitivity analysis. In some cases, external data about ρ may be available (e.g., from previous studies or compendia such as that of Hedges and Hedberg, 2007). It is important to use external values of ρ with considerable caution, because the value of ρ has substantial influence on the results of analyses. In particular, it would be difficult to justify the use of the methods described in this article using estimates of ρ obtained from small samples (small numbers of clusters) because those estimates are likely to be subject to considerable sampling error. Similarly, it would be difficult to justify the use of external estimates of ρ , even from large sample sizes if those estimates were not based on a similar sampling strategy, with similar populations, and similar outcome measures. However, making no correction for the effects of clustering at all corresponds to assuming that $\rho = 0$. The assumption that $\rho = 0$ is often very far from the case, and thus it may introduce more serious biases in the computation of variances than using values of ρ that are slightly in error.

Appendix

Derivation of Sampling Distributions of Effect Size Estimates

The sampling distribution of the effect size estimates proposed in this article all follow from the same theorem, given below.

Theorem

Suppose that $Y \sim N(\mu, a\sigma^2/\tilde{N})$ and that S^2 is a quadratic form in normal variates that is independent of Y , so that $E\{S^2\} = b\sigma^2$ and $V\{S^2\} = 2c\sigma^4$, where a, b, c , and $\tilde{N}$ are known constants. Then

$$T = \sqrt{\frac{\tilde{N}b}{a}} \left(\frac{Y}{S} \right)$$

has approximately the noncentral t distribution with b^2/c degrees of freedom and noncentrality parameter

$$\theta = \sqrt{\frac{\tilde{N}b}{a}} \left(\frac{\mu}{\sigma} \right) = \sqrt{\frac{\tilde{N}b}{a}} \delta,$$

where $\delta = \mu/\sigma$. Consequently,

$$D = \frac{Y\sqrt{b}}{S} = T\sqrt{\frac{a}{\tilde{N}}}$$

is a consistent estimate of the effect size δ with approximate variance

$$\frac{a}{\tilde{N}} + \frac{c\delta^2}{2b}. \quad (\text{A1})$$

An approximately unbiased estimate of δ is given by $DJ(b^2/c)$, where the function $J(x)$ is given by

$$J(x) = 1 - \frac{3}{4x - 1}.$$

Proof

First obtain the approximate sampling distribution of S^2 . Box (1954) gave the approximate sampling distribution of quadratic forms in normal variables (such as S^2 , which is a linear combination of chi-squares) in terms of the first two cumulants of the quadratic form. Theorem 3.1 in Box (1954) implies that S^2 is distributed as approximately a constant g times chi-square with h degrees of freedom, where g and h are given by $g = V\{S^2\}/2E\{S^2\} = c\sigma^2/b$ and $h = 2(E\{S^2\})^2/V\{S^2\} = b^2/c$, where $E\{X\}$ and $V\{X\}$ are the expected value and the variance of X . Therefore, $S^2/gh = S^2/b\sigma^2$ is distributed as a chi-square with h degrees of freedom divided by h . This approximation is generally excellent and is the basis, for example, of the standard tests used in repeated measures analysis of variance (e.g., Geisser & Greenhouse, 1958).

By the definition of the noncentral t distribution (see, e.g., Johnson & Kotz, 1970), it follows that

$$\frac{Y\sqrt{\tilde{N}/a\sigma^2}}{\sqrt{S^2/b\sigma^2}} = \sqrt{\frac{\tilde{N}b}{a}} \left(\frac{Y}{S} \right)$$

has (approximately) the noncentral t distribution with b^2/c degrees of freedom and noncentrality parameter

$$\theta = \sqrt{\frac{\tilde{N}b}{a}} \left(\frac{\mu}{\sigma} \right).$$

Using properties of the noncentral t distribution, via arguments that parallel those in Hedges (1981), it follows that D is a consistent estimator of δ , that

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$DJ(b^2/c)$ is an unbiased estimator of δ , and the variance of D is approximately given by Equation A1.

The theorem can be applied to obtain the sampling distribution of each of the effect size estimators given in this article, using some elementary facts. In each case, apply the theorem with $Y = \bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C$, $\mu = \mu_{\bullet}^T - \mu_{\bullet}^C$, and $\tilde{N} = N^T N^C / (N^T + N^C)$, but with different definitions of S and σ . Therefore, in each case, use the fact that the mean of $\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C$ is given by

$$E\{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C\} = \mu_{\bullet}^T - \mu_{\bullet}^C.$$

However, the variance of $\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C$ and the mean and various choices of S require different derivations in the balanced (equal cluster sample size) and unbalanced (unequal cluster sample size) cases.

Equal Cluster Sample Sizes

In the case of equal cluster sample sizes, a direct argument gives the variance of the mean difference as

$$V\{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C\} = \left(\frac{N^T N^C}{N^T + N^C} \right)^{-1} (\sigma_W^2 + n\sigma_B^2).$$

Also used are the moments of S_B^2 , S_W^2 , and S_T^2 , which are derived from their relation to sums of squares (see, e.g., Snedecor, 1956). Specifically, because $n(M-2)S_B^2/(n\sigma_B^2 + \sigma_W^2)$ has a chi-square distribution with $(M-2)$ degrees of freedom, the mean of S_B^2 is

$$E\{S_B^2\} = \sigma_B^2 + \frac{\sigma_W^2}{n} \quad (\text{A2})$$

and the variance of S_B^2 is

$$V\{S_B^2\} = \frac{2(n\sigma_B^2 + \sigma_W^2)^2}{n^2(M-2)}. \quad (\text{A3})$$

Similarly, because $(N-M)S_W^2/\sigma_W^2$ has the chi-square distribution with $(N-M)$ degrees of freedom, the mean of S_W^2 is σ_W^2 , and the variance of S_W^2 is

$$V\{S_W^2\} = \frac{2\sigma_W^4}{N-M}. \quad (\text{A4})$$

Because

$$S_T^2 = \frac{n(M-2)S_B^2 + (N-M)S_W^2}{N-2},$$

the expected value of S_T^2 follows from the expected values of S_B^2 and S_W^2 , namely,

$$E\{S_T^2\} = \sigma_W^2 + \left(\frac{N-2n}{N-2}\right)\sigma_B^2, \quad (\text{A5})$$

and the variance of S_T^2 follows from the variances of S_B^2 and S_W^2 , namely,

$$V\{S_T^2\} = \frac{2(N-2)\sigma_W^4 + 2n(N-2n)\sigma_B^4 + 4(N-2n)\sigma_B^2\sigma_W^2}{(N-2)^2}. \quad (\text{A6})$$

The Distribution of d_W

In this case, apply the theorem with $\sigma^2 = \sigma_W^2$ and $S^2 = S_W^2$. Here,

$$a = \frac{\sigma_W^2 + n\sigma_B^2}{\sigma_W^2} = \frac{1 + (n-1)\rho}{1 - \rho}.$$

Because the expected value of S_W^2 is σ_W^2 , it follows that $b = 1$. Because the variance of S_W^2 is $2\sigma_W^4/(N-M)$, it follows that $c = 1/(N-M)$. Substituting the expressions for a , b , and c into Equation A1, noting that $\sigma_B^2/\sigma_W^2 = \rho/(1-\rho)$, and simplifying, gives the result in Expression 8. Because S^2 involves only a single chi-square, it follows that the t statistic corresponding to d_W has exactly the noncentral t distribution with $(N-M)$ degrees of freedom.

The Distribution of d_{B1}

In this case, apply the theorem with $\sigma^2 = \sigma_B^2$ and $S^2 = \hat{\sigma}_T^2 = S_B^2 - S_W^2/n$. Here,

$$a = \frac{n\sigma_B^2 + \sigma_W^2}{\sigma_B^2} = \frac{1 + (n-1)\rho}{\rho}.$$

The expected value of S^2 is just σ_B^2 , so $b = 1$. The variance of S^2 is

$$V\{S^2\} = V\{S_B^2\} + \left(\frac{1}{n}\right)^2 V\{S_W^2\} = \frac{2(n\sigma_B^2 + \sigma_W^2)^2}{n^2(M-2)} + \frac{2\sigma_W^4}{n^2(N-M)}$$

so that

$$c = \frac{(n\sigma_B^2 + \sigma_W^2)^2}{n^2(M-2)\sigma_B^4} + \frac{\sigma_W^4}{n^2(N-M)\sigma_B^4}.$$

Substituting the expressions for a , b , and c into Equation A1, noting that $\sigma_W^2/\sigma_B^2 = (1-\rho)/\rho$, and simplifying, gives the result in Expression 10.

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The Distribution of d_{B2}

In this case, apply the theorem with $\sigma^2 = \sigma_B^2$ and $S^2 = S_B^2$. Here, as in d_{B1} ,

$$a = \frac{\sigma_B^2 + \frac{\sigma_W^2}{n}}{\sigma_B^2} = \frac{1 + (n-1)\rho}{\rho}.$$

Using the expected value of S_B^2 given in Equation A2 gives

$$b = \frac{1 + (n-1)\rho}{n\rho}.$$

Using the variance of S_B^2 given in Equation A3 yields

$$c = \frac{(n\sigma_B^2 + \sigma_W^2)^2}{n^2(M-2)\sigma_B^2}.$$

Substituting the expressions for a , b , and c into Equation A1, noting that $\sigma_W^2/\sigma_B^2 = (1-\rho)/\rho$, and simplifying, gives the result in Expression 12. Because S^2 involves only a single chi-square, it follows that the t statistic corresponding to d_{B2} has exactly the noncentral t distribution with $(M-2)$ degrees of freedom.

The Distribution of d_{T1}

In this case, apply the theorem with $\sigma^2 = \sigma_T^2 = \sigma_W^2 + \sigma_B^2$ and $S^2 = \hat{\sigma}_T^2$. Here,

$$a = \frac{\sigma_W^2 + n\sigma_B^2}{\sigma_W^2 + \sigma_B^2} = 1 + (n-1)\rho.$$

The expected value of S^2 is just σ_T^2 , so $b = 1$. The variance of S^2 is

$$V\{S^2\} = V\{S_B^2\} + \left(\frac{n-1}{n}\right)^2 V\{S_W^2\} = \frac{2(n\sigma_B^2 + \sigma_W^2)^2}{n^2(M-2)} + \left(\frac{n-1}{n}\right)^2 \left(\frac{2\sigma_W^4}{N-M}\right)$$

so that

$$c = \frac{2(n\sigma_B^2 + \sigma_W^2)^2}{n^2(M-2)(\sigma_B^2 + \sigma_W^2)^2} + \left(\frac{n-1}{n}\right)^2 \left(\frac{2\sigma_W^4}{(N-M)(\sigma_B^2 + \sigma_W^2)^2}\right).$$

Substituting the expressions for a , b , and c into Equation A1, noting that $\rho = \sigma_B^2/(\sigma_B^2 + \sigma_W^2)$ and $1-\rho = \sigma_W^2/(\sigma_B^2 + \sigma_W^2)$, and simplifying, gives the result in Expression 14.

The Distribution of d_{T2}

In this case, apply the theorem with $\sigma^2 = \sigma_T^2 = \sigma_W^2 + \sigma_B^2$ and $S^2 = S_T^2$. Here, as in d_{T1} ,

$$a = \frac{\sigma_W^2 + n\sigma_B^2}{\sigma_W^2 + \sigma_B^2} = 1 + (n-1)\rho.$$

Using the expected value of S_T^2 given in Equation A5, we compute

$$b = \frac{\sigma_W^2 + \left(\frac{N-2n}{N-2}\right)\sigma_B^2}{\sigma_W^2 + \sigma_B^2} = 1 - \frac{2(n-1)\rho}{N-2}.$$

Using the variance of S_T^2 given in Equation A6, compute

$$c = \frac{(N-2)\sigma_W^4 + n(N-2n)\sigma_B^4 + 2(N-2n)\sigma_B^2\sigma_W^2}{(N-2)^2(\sigma_W^2 + \sigma_B^2)^2}.$$

Substituting the expressions for a , b , and c into Equation A1, dividing the numerator and denominator by σ_W^2 , noting that $\rho = \sigma_B^2/(\sigma_B^2 + \sigma_W^2)$ and $1 - \rho = \sigma_W^2/(\sigma_B^2 + \sigma_W^2)$, and simplifying, gives the result in Expression 16.

Unequal Cluster Sample Sizes

When cluster sample sizes are unequal, expressions for the effect size estimators and their variances are more complex. First derive the variance of the mean differences. A direct argument leads to

$$V\{\bar{Y}_{\bullet\bullet}^T - \bar{Y}_{\bullet\bullet}^C\} = \left(\frac{N^T N^C}{N^T + N^C}\right)^{-1} (\sigma_W^2 + \tilde{n}\sigma_B^2) \quad (\text{A7})$$

and

$$V\{\bar{Y}_{*\bullet}^T - \bar{Y}_{*\bullet}^C\} = \left(\frac{m^T m^C}{m^T + m^C}\right)^{-1} (\sigma_B^2 + \bar{n}_B\sigma_B^2), \quad (\text{A8})$$

where $\tilde{n}$ and $\bar{n}_B$ are defined in the text. The expected value and variance of S_T^2 can be calculated from the analysis of variance across clusters within the treatment groups. When cluster sample sizes are unequal, the between- and within-cluster sums of squares are still independent, and the within-cluster sum of squares has a chi-square distribution, but if $\rho \neq 1$, the between-cluster sum of squares does not have a chi-square distribution. However, because the between-cluster sum of squares is a quadratic form, the methods used in this article apply,

and the distribution of effect size estimates can be obtained. To obtain the expected value of S_T^2 , use the fact that

$$S_T^2 = \frac{SSB^T + SSW^T + SSB^C + SSW^C}{N - 2},$$

where SSB^T and SSW^T and SSB^C and SSW^C are the sums of squares between and within clusters in the treatment and control groups, respectively. Using the expected values of the $SSBs$ and $SSWs$ given, for example, in Equations 77 and 78 on page 70 of Searle, Casella, and McCulloch (1992) obtains

$$E\{S_T^2\} = \sigma_w^2 + \frac{B\sigma_B^2}{N - 2}, \quad (A9)$$

where B is the auxiliary constant defined in Equation 20. Because the between-cluster variance component estimates in the treatment and control groups are

$$(\hat{\sigma}_B^T)^2 = \frac{MSB^T - MSW^T}{n_U^T}$$

and

$$(\hat{\sigma}_B^C)^2 = \frac{MSB^C - MSW^C}{n_U^C},$$

it follows that S_T^2 can be written as a function of between- and within-cluster variance components

$$S_T^2 = \frac{SSW^T + SSW^C + A^T (\hat{\sigma}_B^T)^2 + A^C (\hat{\sigma}_B^C)^2}{N - 2}.$$

Therefore, the variance of S_T^2 is given by

$$\begin{aligned} (N - 2)^2 V\{S_T^2\} &= V\{SSW^T\} + V\{SSW^C\} + (A^T)^2 V\{(\hat{\sigma}_B^T)^2\} + (A^C)^2 V\{(\hat{\sigma}_B^C)^2\} \\ &\quad + 2\text{Cov}(SSW^T, (\hat{\sigma}_B^T)^2) + 2\text{Cov}(SSW^C, (\hat{\sigma}_B^C)^2). \end{aligned}$$

Using Expressions 95 and 102 for the variances of the sums of squares and the variance component estimates and Expression 96 for the covariance term from pages 74 and 75 of Searle et al. (1992), and simplifying yields

$$V\{S_T^2\} = \frac{2\sigma_w^4}{N - 2} + \frac{2B\sigma_B^2\sigma_w^2}{(N - 2)^2} + \frac{2A\sigma_B^4}{(n - 2)^2}, \quad (A10)$$

where A and B are the auxiliary constants in Equation 20.

The expected value and variance of S_B^2 can be derived directly. Writing $S_B^2 = (\mathbf{y}_T' \mathbf{A}_T \mathbf{y}_T + \mathbf{y}_C' \mathbf{A}_C \mathbf{y}_C) / (M - 2)$, where $\mathbf{y}_T$ and $\mathbf{y}_C$ are vectors of treatment and control group cluster means, respectively, and $\mathbf{A}_T$ and $\mathbf{A}_C$ are m^T by m^T and m^C by m^C matrices defined by $\mathbf{A}_T = \mathbf{I} - \mathbf{1}\mathbf{1}'/m^T$ and $\mathbf{A}_C = \mathbf{I} - \mathbf{1}\mathbf{1}'/m^C$, respectively, where $\mathbf{I}$ is an identity matrix and $\mathbf{1}$ is a column vector of 1's of appropriate dimensions. A direct but tedious application of a theorem on the mean and variance of quadratic forms in normal variables (see, e.g., Searle, 1971, p. 57) gives the mean of S_B^2 as $\{\text{trace}(\mathbf{A}_T \mathbf{V}_T) + \text{trace}(\mathbf{A}_C \mathbf{V}_C)\} / (M - 2)$ and variance of S_B^2 as $2\{\text{trace}(\mathbf{A}_T \mathbf{V}_T \mathbf{A}_T \mathbf{V}_T) + \text{trace}(\mathbf{A}_C \mathbf{V}_C \mathbf{A}_C \mathbf{V}_C)\} / (M - 2)^2$, where $\mathbf{V}_T$ and $\mathbf{V}_C$ are the covariance matrices of $\mathbf{y}_T$ and $\mathbf{y}_C$. Here $\mathbf{V}_T$ is an $m^T \times m^T$ diagonal matrix whose i th diagonal element is $\sigma_B^2 + \sigma_W^2/n_i^T$, and $\mathbf{V}_C$ is an $m^C \times m^C$ diagonal matrix whose i th diagonal element is $\sigma_B^2 + \sigma_W^2/n_i^C$. Using this theorem obtains

$$E\{S_B^2\} = \sigma_B^2 + \frac{\sigma_W^2}{\bar{n}_B} \quad (\text{A11})$$

and

$$V\{S_B^2\} = \frac{2(M - 2)\sigma_B^4 + 2C_1\sigma_W^4 + 4C_2\sigma_B^2\sigma_W^2}{(M - 2)^2}, \quad (\text{A12})$$

where $C_1 = (m^T - 2)\bar{n}_I^{T2} + (m^C - 2)\bar{n}_I^{C2} + (\bar{n}_I^T)^2 + (\bar{n}_I^C)^2$ and $C_2 = (m^T - 1)\bar{n}_I^T + (m^C - 1)\bar{n}_I^C$.

The Distribution of d_W

To obtain the distribution of d_W , apply the theorem with $\sigma^2 = \sigma_W^2$ and $S^2 = S_W^2$. Here,

$$a = \frac{\sigma_W^2 + \tilde{n}\sigma_B^2}{\sigma_W^2} = \frac{1 + (\tilde{n} - 1)\rho}{1 - \rho},$$

$b = E\{S_W^2\}/\sigma_W^2 = 1$, and $c = V\{S_W^2\}/2\sigma_W^2 = 1/(N - M)$. Substituting the expressions for a , b , and c into Equation A1 and simplifying gives Equation 18. Note that because S^2 involves only a single chi-square, it follows that the t statistic corresponding to d_W has exactly the noncentral t distribution with $(N - M)$ degrees of freedom.

The Distribution of d_T

In this case, apply the theorem with $\sigma^2 = \sigma_T^2 = \sigma_B^2 + \sigma_W^2$ and $S^2 = S_T^2$. Here,

$$a = \frac{\sigma_W^2 + \tilde{n}\sigma_B^2}{\sigma_B^2 + \sigma_W^2} = 1 + (\tilde{n} - 1)\rho.$$

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Using the expected value of S_T^2 given in Equation A9, compute

$$b = \frac{\sigma_W^2 + \frac{B\sigma_B^2}{N-2}}{\sigma_W^2 + \sigma_B^2} = 1 - \rho \left(\frac{N-2-B}{N-2} \right).$$

Using the variance of S_T^2 given in Equation A10, compute

$$c = \frac{(N-2)\sigma_W^4 + A\sigma_B^4 + 2B\sigma_B^2\sigma_W^2}{(N-2)^2(\sigma_W^2 + \sigma_B^2)^2} = \frac{(N-2)(1-\rho)^2 + A\rho^2 + 2B\rho(1-\rho)}{(N-2)^2}.$$

Substituting the expressions for a , b , and c into Equation A1 and simplifying gives the result in Expression 20.

The Distribution of d_B

To obtain the distribution of d_B , apply the theorem with $\sigma^2 = \sigma_B^2$ and $S^2 = S_B^2$. Here,

$$a = \frac{\sigma_W^2 + \bar{n}_B\sigma_B^2}{\sigma_B^2} = \frac{1 + (\bar{n}_B - 1)\rho}{\rho}.$$

Using the expected value of S_B^2 given in Equation A11 gives

$$b = \frac{\sigma_B^2 + \frac{\sigma_W^2}{\bar{n}_B}}{\sigma_B^2} = \frac{1 + (\bar{n}_B - 1)\rho}{\bar{n}_B\rho}.$$

Using the variance of S_B^2 given in Equation A12, compute $c = V\{S_B^2\}/2\sigma_B^4$ as

$$c = \frac{(M-2)\sigma_B^4 + C_1\sigma_W^4 + 2C_2\sigma_B^2\sigma_W^2}{(M-2)^2\sigma_B^4} = \frac{(M-2)\rho^2 + C_1(1-\rho)^2 + 2C_2\rho(1-\rho)}{(M-2)^2},$$

where C_1 and C_2 are given above. Substituting a , b , and c into Equation A1 and simplifying yields Equation 22.

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